

IT-SIMPLERA INSTITUTE

Cyber Security Analyst Internship Program

WEEK 02 ASSIGNMENT

Computer Networking Fundamentals & Network Configuration

Prepared By

TALAL REHAN

Assigned To

Cyber Security Analyst Interns

Assignment Overview

This assignment is designed to introduce interns to the fundamental concepts of computer networking and network communication. Through practical exercises and theoretical comparisons, interns will learn how to identify network configuration details, use essential networking commands, understand the differences between routers and switches, compare IPv4 and IPv6 addressing, and explore the seven layers of the OSI Model. The assignment aims to strengthen both practical networking skills and theoretical knowledge, providing a solid foundation for future studies in cybersecurity and network administration.

Submission Details

Week: 01

Submission Deadline: 10/JULY/2026

Submission Format: PDF Report + Network Diagram + Screenshots (if applicable)

Evaluation Criteria

Documentation:10, LinkedIn:50, GitHub:10, Interview:30

Company Information

Company: IT-SIMPLERA Institute

Department: Cyber Security

Supervisor: Senior Cyber Security Analyst

Objectives

- ☐ Understand the basic concepts of computer networking and network communication.
- ☐ Learn how to identify and interpret network configuration using the **ipconfig /all** command.
- ☐ Gain hands-on experience with essential networking commands such as **ping**, **tracert**, **nslookup**, and **arp**.
- ☐ Differentiate between the functions and use cases of routers and switches.
- ☐ Compare the features and advantages of IPv4 and IPv6 addressing.
- ☐ Understand the seven layers of the OSI Model, their functions, associated protocols, and networking devices.
- ☐ Develop practical troubleshooting skills and improve technical documentation through screenshots and written explanations.

Tasks

Network Information

Q1. Using the ipconfig /all command, identify your computer's IPv4 Address, MAC Address, and Default Gateway. Explain the purpose of each and attach screenshots of your results.

Basic Networking Commands

Q2. Execute the following networking commands on your computer. Attach screenshots of the output and write 1–2 lines explaining the purpose of each command.

```
ipconfig /all  
ping google.com  
tracert google.com  
nslookup google.com  
arp -a
```

Router vs. Switch

Q3. Compare a Router and a Switch by creating a table with at least six comparison points, including their functions, OSI layer, addressing, and common use cases.

IPv4 vs. IPv6

Q4. Compare IPv4 and IPv6 by listing at least six key differences in a table. Include one valid example of an IPv4 address and one valid example of an IPv6 address.

OSI Model

Q5. List all seven layers of the OSI Model in the correct order. For each layer, write: Its primary function One commonly used protocol One networking device associated with that layer

Tools Required

Microsoft Word, PowerPoint/Draw.io/Visio, Internet Browser, PDF Converter

Report Structure

Cover Page, TOC, Introduction, Tasks 1-5, Challenges, Conclusion, References

Submission Format

Week2 folder with Report.pdf, Diagram, Screenshots, LinkedIn link, GitHub link.

Learning Outcomes

By the end of this assignment, interns will be able to:

- Identify and interpret key network configuration details, including IPv4 address, MAC address, and Default Gateway.
- Use common networking commands to diagnose and troubleshoot basic network connectivity issues.
- Explain the purpose and functionality of networking commands such as **ipconfig**, **ping**, **tracert**, **nslookup**, and **arp**.
- Compare the functions, characteristics, and use cases of routers and switches.
- Differentiate between IPv4 and IPv6 based on their addressing schemes, features, and capabilities.
- Describe the seven layers of the OSI Model, including their primary functions, common protocols, and associated networking devices.
- Improve practical networking skills, technical documentation, and problem-solving abilities through hands-on exercises and analysis.

Submission Deadline: 10/July/2026